Single VO_2 Neuristor Simulations

Model, specimen parameters, and loss-function comparison

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Goal: Model Specimen Behavior Across Current Inputs

Experimental target

Use the supplied experimental data to reproduce the specimen's response to the 22 input currents studied in the paper.

1. Preserve the published electrothermal and hysteresis equations.
2. Set specimen parameters from the paper and supplied measurements.
3. Compare voltage level, amplitude, and frequency.

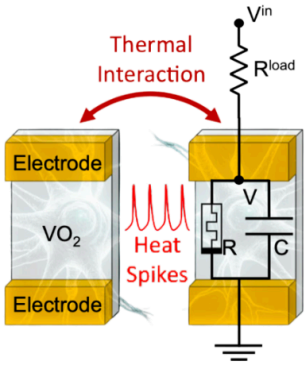
Experimental context: Gildor et al., arXiv:2604.04594v1. Numerical inputs: supplied traces and R(T) data.

Starting Point: Electrical Dynamics

Yuanhang voltage-source node equation

$$C\frac{dV}{dt} = \frac{V_{in} - V}{R_{series}} - \frac{V}{R_{\mathrm{VO_2}}}$$

- $C\,dV/dt$: charge stored at the neuristor node.
- $(V_{in} - V)/R_{series}$: current delivered by the source.
- $V/R_{\mathrm{VO_2}}$: nonlinear leakage through VO_2.
- $R_{\mathrm{VO_2}} = R(T, \mathcal{H})$: temperature and memory couple the electrical and thermal problems.



Voltage-source equivalent circuit.

Starting point: Y.-H. Zhang et al., Nature Communications (2024).

The Feedback Mechanism: Thermal Dynamics

Single-neuristor reduction

$$C_{th}\frac{dT}{dt} = \underbrace{\frac{V^2}{R_{\mathrm{VO_2}}}}_{\text{Joule heating}} - \underbrace{S_e(T-T_0)}_{\text{cooling}} + \underbrace{\sigma\eta(t)}_{\text{optional noise}}$$

- C_{th}: thermal inertia of the active device.
- S_e: thermal conductance to substrate/environment.
- Heating lowers $R_{\mathrm{VO_2}}$; lower resistance changes the voltage and power.
- This delayed negative feedback enables relaxation oscillations.

For one isolated device, the inter-neuristor thermal coupling term is zero.

Two coupled state variables

$$V(t), \qquad T(t)$$

Material memory

$$\mathcal{H} = \{\delta, T_r, g_r, T_{pr}, \ldots\}$$

Electrical storage
⇕
Joule heating
⇕
Hysteretic switching

Material Law: Resistance and Hysteresis

$$R(T,\mathcal{H}) = R_0 e^{E_a/(k_B T)} \underbrace{g(T,\mathcal{H})}_{\text{insulating fraction}} + R_m$$

$$g = \frac{1}{2} + \frac{1}{2}\tanh\left[\beta\left(\delta\frac{w}{2} + T_c - (T+T_p)\right)\right]$$

- T_c, w, β: center, width, and steepness.
- $\delta = +1$: heating; $\delta = -1$: cooling.
- T_p: minor-loop shift after a reversal.
- The same law is used in both source configurations.

Hysteresis: L. A. L. de Almeida et al., Optical Engineering (2002).

Insulating and metallic limits

$g \approx 1$: insulating Arrhenius contribution.

$g \approx 0$: metallic floor $R \approx R_m$.

History matters

$T_p = 0$ before the first reversal. After a reversal, the stored state shifts the effective temperature to form a minor branch.

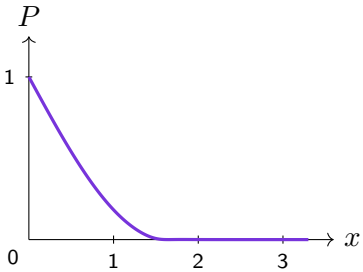
Minor Loops: The Proximity Function

A reversal shifts the effective temperature

$$T_{\text{eff}} = T + T_p, \qquad T_p = m\,T_{pr}\,P(x;\gamma), \qquad x = \frac{T - T_r}{T_{pr} + \varepsilon}.$$

$$\boxed{P(x;\gamma) = \frac{1 - \sin(\gamma x)}{2}\left[1 + \tanh(\pi^2 - 2\pi x)\right]}$$

- T_r: latest detected reversal temperature; T_{pr}: signed shift in K.
- $m = 0$ before the first reversal; $m = 1$ afterward.
- Near $x = 0$, $P \approx 1$: the shift preserves the reversal state.
- As $x \to +\infty$, $P \to 0$: the selected major branch is recovered.



Default $\gamma = 0.95627$; dimensionless $x \geq 0$ shown.

Role of γ

Controls the minor-loop shape. Major-loop data alone do not identify it.

Default Yuanhang/Almeida law: `src/neuristor/model.py`. Implementation uses $\varepsilon = 10^{-6}$ K and float32 arithmetic.

Minor Loops: Set the Shift at Each Reversal

1. Save the state, then change direction

$$T_r \leftarrow T, \quad g_r \leftarrow g(T, \mathcal{H}^-), \quad \delta \leftarrow \operatorname{sign}(T - T^{\text{last}}).$$

$\mathcal{H}^-$ is the memory before changing the branch.

2. Match the fraction at the reversal

$$g_r \simeq \frac{1}{2} + \frac{1}{2}\tanh\Big[\beta\left(\delta\frac{w}{2} + T_c - T_r - T_{pr}\right)\Big].$$

Inverting the tanh gives the stored shift:

$$\boxed{T_{pr} = \delta\frac{w}{2} + T_c - T_r - \frac{\operatorname{arctanh}(2g_r - 1)}{\beta}.}$$

3. Follow the resulting minor branch

$$g(T, \mathcal{H}) = \frac{1}{2} + \frac{1}{2}\tanh\Big[\beta\left(\delta\frac{w}{2} + T_c - T - T_p\right)\Big].$$

Why this carries memory

T_r, g_r set the starting point; $T_{pr}P$ carries that state into the new direction. Each new reversal replaces the stored anchor and shift.

Detection in the solver

Accept motion if $|T - T^{\text{last}}| > 0.01$ K. Reverse if its sign differs from δ. Smaller motion accumulates from the last accepted temperature.

`model.py:HysteresisArray`: matching uses $P(0) \approx 1$. Only exact float32 endpoints $g_r = 0, 1$ are regularized for arctanh.

From Voltage Drive to Ideal Current Drive

Original voltage source

$$C\frac{dV}{dt} = \frac{V_{in} - V}{R_{series}} - \frac{V}{R}$$

I_{src} changes with V

Ideal imposed current

$$I_{in} = C\frac{dV}{dt} + \frac{V}{R}$$

$$\boxed{C\frac{dV}{dt} = I_{in} - \frac{V}{R}}$$

Preserved	**Assumed in the current-source reduction**
Yuanhang $R(T, \mathcal{H})$	one VO_2 device/domain
single thermal state T	one thermal node
Joule heating and cooling	ideal imposed current
deterministic replay ($\sigma = 0$)	no added phase-lag state

$C = 0$: $V = I_{in}R$. A series resistor does not change an ideal imposed current.

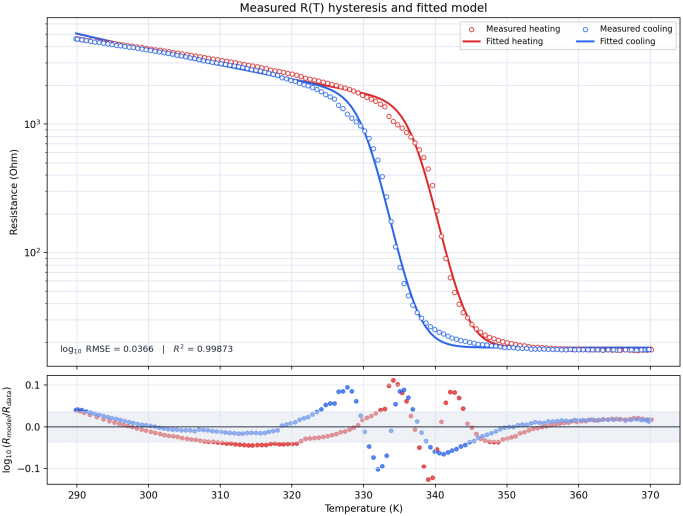
Specimen R(T): Fit the Measured Major Loop

$$\min_{\theta} \sum_{i} [\log_{10} R_{\mathrm{model}}(T_i;\theta) - \log_{10} R_i]^2$$

Parameter	Specimen	Yuanhang
R_0 (Ω)	0.7989	0.00536
E_a/k_B (K)	2536.97	5220.47
R_m (Ω)	18.215	1286.32
T_c (K)	333.494	332.806
w (K)	6.882	7.194
β (K^{-1})	0.2992	0.2528

Fit quality

$R^2 = 0.99873$ $\quad \mathrm{RMSE}_{\log_{10} R} = 0.03663$



264 measured points from sample 100425 chip1 gap3; reviewed fit bundle 0849a9.

Specimen Parameters: Fitted Values and Unknowns

Parameter	Specimen value	Basis
R_0	0.7989 Ω	Fitted to measured R(T)
E_a/k_B	2536.97 K	Fitted to measured R(T)
R_m	18.215 Ω	Fitted to measured R(T)
T_c	333.494 K	Fitted to measured R(T)
w	6.882 K	Fitted to measured R(T)
β	0.2992 K^{-1}	Fitted to measured R(T)
T_0	314.4 K	Adopted from paper temperatures*
Parameters still to determine for this specimen		
γ	Unknown	Minor-loop shape
S_e	Unknown	Thermal conductance to the environment
C	Unknown	Electrical capacitance
C_{th}	Unknown	Thermal capacitance

*Paper temperatures: 41.1 and 41.4 °C (314.25–314.55 K); midpoint adopted.

Noise strength is fixed at $\sigma = 0$.

R(T) fit: bundle 0849a9. Ambient temperature: Gildor et al., arXiv:2604.04594v1, pp. 3, 5.

From Model Parameters to Frequency and Amplitude

$$\theta = (\gamma, S_e, C, C_{th}), \qquad V_i^{\mathrm{sim}}(t;\theta) = \mathcal{S}[I_i^{\mathrm{meas}}(t);\theta].$$

$\mathcal{S}$ solves the electrical, thermal, and hysteresis equations. The six R(T) parameters and T_0 remain fixed; i labels an input-current record.

Measure complete voltage cycles in the 150–250 ns window

$$f_i(\theta) = \frac{1000}{\mathrm{median}_k\, P_{ik}\ [\mathrm{ns}]}\ \mathrm{MHz}, \qquad A_i(\theta) = \mathrm{median}_k\, H_{ik},$$

$$H_{ik} = \frac{V_{ik}^{\mathrm{peak}} + V_{i,k+1}^{\mathrm{peak}}}{2} - V_{ik}^{\mathrm{trough}}.$$

Vpp means peak-to-peak voltage amplitude: the full height from a trough to a peak, not half that height. For each complete cycle k, P_{ik} is the time between consecutive peaks; H_{ik} uses their average height above the intervening trough.

Apply the same measurements to the experimental voltage to obtain f_i^{meas} and A_i^{meas}. Compare them with the simulation to calculate the loss.

Same acquisition-grid peak detection for measured and simulated traces. Missing complete cycles leave frequency undefined.

Loss Functions: Frequency Only and Frequency + Amplitude

For each of the 11 selected oscillatory records, define the relative errors:

$$e_{f,i}(\theta) = \frac{f_i(\theta) - f_i^{\text{meas}}}{f_i^{\text{meas}}}, \qquad e_{A,i}(\theta) = \frac{A_i(\theta) - A_i^{\text{meas}}}{A_i^{\text{meas}}}.$$

Fit 1: frequency only

$$L_f(\theta) = \frac{1}{11}\sum_{i=1}^{11} e_{f,i}(\theta)^2.$$

Minimize the average squared relative frequency error. Check amplitude and mean voltage afterward.

Fit 2: frequency and amplitude

$$L_{f,A}(\theta) = L_f(\theta) + \frac{1}{11}\sum_{i=1}^{11} e_{A,i}(\theta)^2.$$

Add the average squared relative Vpp error. Equal percentage errors have equal weight.

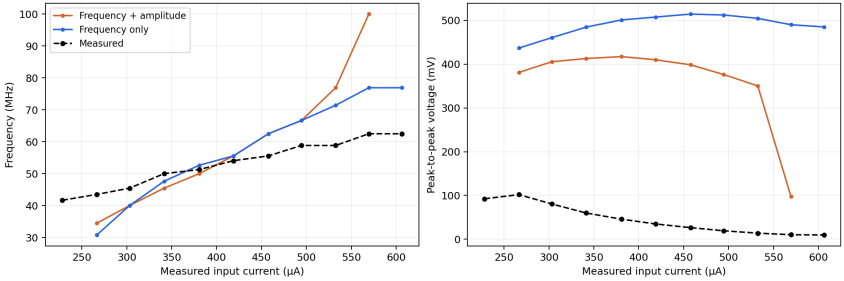
A 10% error contributes $0.10^2 = 0.01$; an exact match contributes zero. The optimizer changes θ, reruns the simulator and seeks a smaller loss using one parameter set for all 11 records.

Same fixed parameters, bounds, starts and budget. Fit 11 inputs (228–606 μA); replay all 22 at three timesteps.

Separate checks: complete sustained cycles, $f \leq 110$ MHz, temperature range and proximity response.

Paired frequency-only and frequency-plus-amplitude recipes, 20 October 2026. Search bounds are choices, not measured intervals.

New Results: Frequency and Amplitude Across the Inputs



Search	Passing targets	Frequency error	Vpp error
Frequency only	10/11	14.5% (10/11)	1965.7%
Frequency + amplitude	8/11	18.2% (9/11)	903.0%

Errors are mean absolute percentage errors. Frequency uses only records with a defined frequency (count shown); Vpp uses all 11 targets.

Neither is an all-target fit; the combined candidate drops from 9 to 8 passing targets at the finest step.

▸ Open all 44 GIFs: measured current, voltage comparison and R(T)

New runs: 543408 and 5ad999. Finest replay: 0.00625 ns. Missing frequencies are gaps, not zero-frequency predictions.

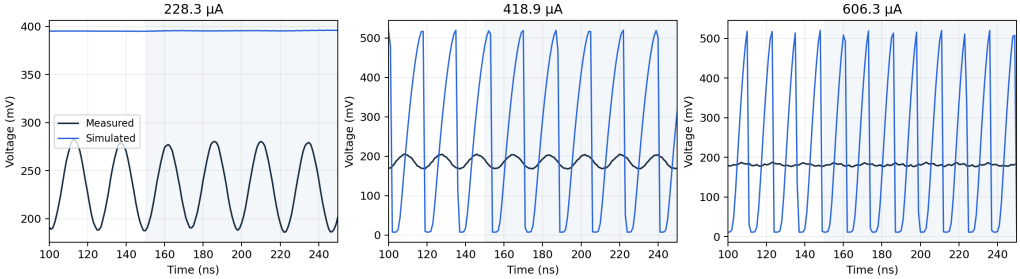
Parameter Values Selected by Each Search

Parameter	Frequency only	Frequency + amplitude
γ	0.7075	0.6308
S_e (μW/K)	5.794	4.096
C (pF)	5.361	6.254
C_{th} (pJ/K)	0.02800	0.03007

The six R(T) values and $T_0 = 314.4$ K were fixed. These are search candidates, not established specimen values.

Shared seed, starting points and evaluation cap; 250 and 250 unique evaluations. Comparison bundle 26574a.

Frequency only: Three Representative Measured Inputs

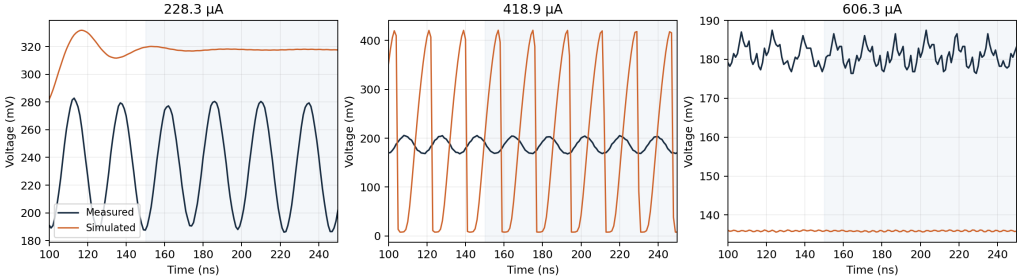


Input (μA)	Frequency (MHz)		Vpp (mV)		Cycle/frequency check
	Measured	Simulated	Measured	Simulated	
228.3	41.7	–	92.4	–	Fail
418.9	54.1	55.6	34.8	507.8	Pass
606.3	62.5	76.9	9.6	485.1	Pass

Examples selected by current rank. Shading: 150–250 ns loss window. No phase alignment. A dash means no detected complete-cycle measurement.

Measured current: 50–250 ns mean after baseline subtraction. Finest replay: 0.00625 ns.

Frequency + amplitude: Three Representative Measured Inputs



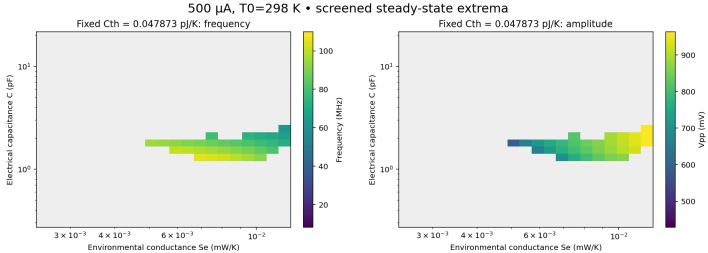
Input (μA)	Frequency (MHz)		Vpp (mV)		Cycle/frequency check
	Measured	Simulated	Measured	Simulated	
228.3	41.7	–	92.4	–	Fail
418.9	54.1	55.6	34.8	410.0	Pass
606.3	62.5	–	9.6	–	Fail

Examples selected by current rank. Shading: 150–250 ns loss window. No phase alignment. A dash means no detected complete-cycle measurement.

Measured current: 50–250 ns mean after baseline subtraction. Finest replay: 0.00625 ns.

Exploring How the Three Parameters Change the Response

The fits did not reproduce the measured frequency and amplitude together. To better understand their influence, we explored C, C_{th} and S_e across a parameter grid.



Example slice: vary C and S_e at fixed C_{th}.

Heatmaps show frequency and peak-to-peak voltage. Sliders fix one parameter while the other two vary; click a retained point to see its voltage and R(T) animation.

▸ Open the three-parameter heatmaps and GIFs

12,167 combinations; 1,258 original retained points. Fixed 500 μA, 298 K, $\gamma = 0.95627$ and fitted R(T). These differ from the experimental replay conditions.

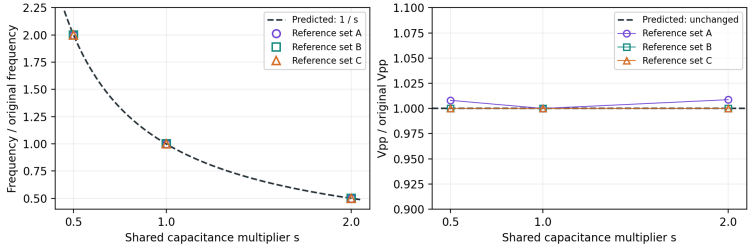
Gray cells are excluded/unresolved, not physical boundaries. Viewer 4325d1; source eaf6ab.

A Sparse Follow-up: Capacitance and Oscillation Timing

To isolate the effect, test three reference parameter sets and multiply both capacitances by $s = 0.5,\ 1,\ 2$, keeping S_e and the operating conditions fixed.

$$(C, C_{th}) \longrightarrow (sC, sC_{th}) \quad \Rightarrow \quad V_s(t) = V(t/s), \quad T_s(t) = T(t/s),$$

$$\boxed{f_s = f/s, \qquad V_{pp,s} = V_{pp}.}$$

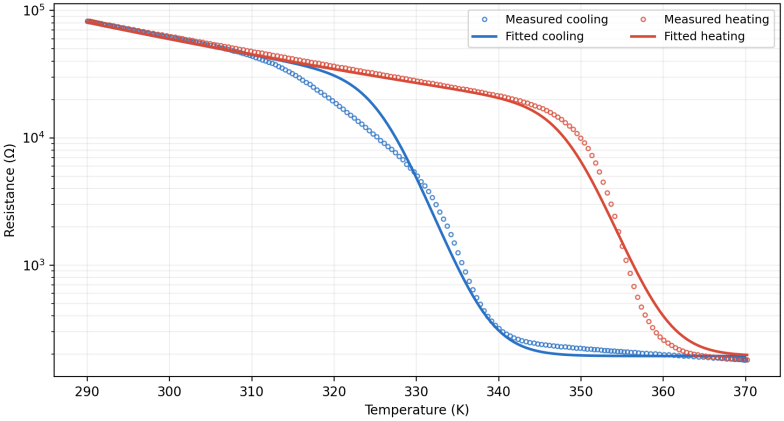


Double both capacitances: half the frequency, with the same voltage–temperature path. Changing C or C_{th} individually can also change the response shape; their amplitude effects need not have the same direction.

Constant ideal current 500 μA, 298 K, $\gamma = 0.95627$. Follow-up 3599fb: 12/12 scale checks pass 2%; plotted points use co-scaled timesteps. Extracted Vpp varies by less than 0.9% because the feature measurement uses a fixed-time smoothing rule.

A Second Specimen: Mica

Mica is another specimen studied by Amir for which we want a model (sample M2, 1600 nm).



Fitted parameter	Mica
R_0 (Ω)	9.415
E_a/k_B (K)	2624.40
R_m (Ω)	193.54
T_c (K)	337.139
w (K)	22.815
β (K^{-1})	0.18294

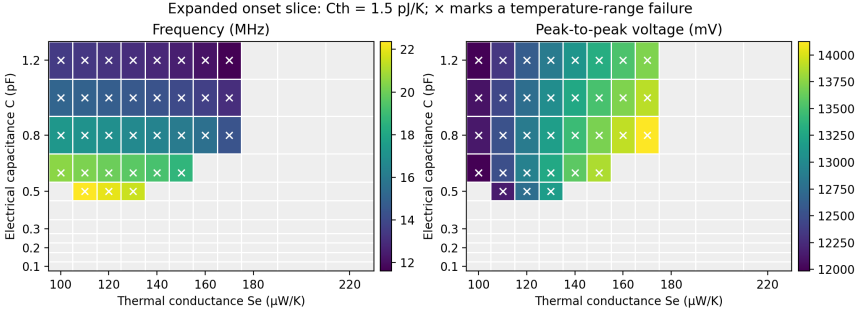
Adopt $T_0 = T_{\text{initial}} = 298$ K, matching the parameter explorations.

The static fit is approximate: $\log_{10}$ RMSE $= 0.08065$, with systematic transition errors.

374 measured R(T) points, 289.98–370.18 K; review c7387b. The usual static-fit cutoff is 0.05. No specimen calibration is adopted.

Mica: Exploring the Parameter Domain

Using Mica's fitted R(T), vary C, C_{th} and S_e to examine their effects on frequency and peak-to-peak voltage, as for the first specimen.



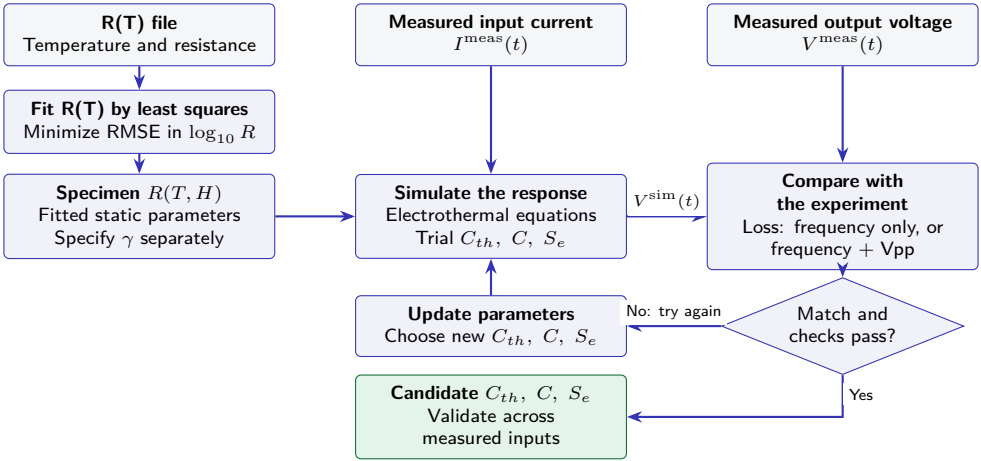
Expanded exploration: 1,843 additional grid entries; 2,616 in total, including overlaps. No point passed every check. All detected sustained oscillations still exceeded the measured R(T) temperature range.

▸ Open Mica's latest completed parameter exploration

Expanded review 226099. Fixed 500 μA, 298 K and assumed $\gamma = 0.95627$. Crosses are excluded diagnostic results; gray cells have no sustained-cycle value.

This search does not establish physical boundaries.

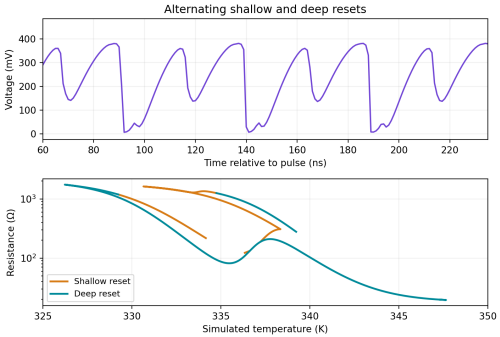
From Measured Data to Specimen Parameters



***** We also explored the domain of C_{th}, C, S_e to visualize their effects on frequency and amplitude.

The fitted dynamic values remain candidates until validated. The static R(T) fit alone does not determine γ; earlier fitting trials also varied it.

Alternating Reset Depths in One Simulated Run



Source label 400 mV (about 303 μA).
Successive voltage swings: about 240 and 350 mV Vpp.

▸ Open the 44 current / voltage / R(T) GIFs

Two relevant time scales

$$\tau_{th} = \frac{C_{th}}{S_e} = 11.91 \text{ ns}$$

$$\tau_e(t) = CR(T, \mathcal{H}) \simeq 0.13\text{–}12 \text{ ns}$$

Along the displayed replay. Fully metallic limit: $CR_m = 0.125$ ns.

Working explanation

RC recharge competes with thermal recovery. Hysteresis memory could alternate partial (about 338 K) and deep (about 348 K) resets.

$C = 6.835$ pF, $C_{th} = 0.04421$ pJ/K,
$S_e = 3.713$ μW/K, $\gamma = 0.15696$.

Numerical limit: short-window alternation survives refinement. Longer constant-current runs remain timestep-sensitive. Stable period doubling is unproven.

Historical diagnostic at 314.363 K, original specimen. Source 717797; audit 1072ef; GIFs 4b9e8c. The causal mechanism remains a hypothesis.